Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Which of the following statements is/are wrong -

(A) van der Waal's radius of iodine is more than its covalent radius

(B) All isoelectronic ions belong to same period of the periodic table.

(C) I.E. $_1$ of N is higher than that of O while I.E $_2$ of O is higher than that of N

(D) The electron gain enthalpy of N is almost zero while that of P is $74.3\ \mathrm{kJ\ mol^{-1}}$

Q.2

The ground state electronic configurations of the elements, U, V, W, X, and Y (these symbols do not have any chemical significance) are as follows :

U $1s^2 2s^2 2p^3$

V $1\ s^2 2\ s^2 2p^6 3\ s^1$

W $1\ s^2 2\ s^2 2p^6 3\ s^2 3p^2$

X $1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6 3\ d^5 4\ s^2$

Y $1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6 3\ d^{10} 4\ s^2 4p^6$

Determine which sequence of elements satisfy the following statements -

(i) Element forms a carbonate which is not decomposed by heating

(ii) Element is most likely to form coloured ionic compounds

(iii) Element has largest atomic radius

(iv) Element forms only acidic oxide

(A) VWYU

(B) VXYW

(C) VWYX

(D) VXWU

Q.3

If the ionization enthalpy and electron gain enthalpy of an elements are 275 and $86\mathrm{kcalmol^{-1}}$ respectively, then the electronegativity of the element on the Mulliken scale is -

(A) 2.8

(B) 0

(C) 4.0

(D) 2.6

@IITJEE_Advanced

Q.4

True position of lanthanides are -

(A) after III B group and in the 6th period

(B) after III B group in the 3rd period

(C) after VI B group in the 3rd period

(D) after VI B group and in the 6th period

Q.5

The screening effect of inner electrons of the nucleus causes

(A) A decrease in the ionization potential

(B) An increase in the ionization potential

(C) No effect on the ionization potential

(D) An increase in the attraction of the nucleus to the electrons

Q.6

The correct order of non-metallic character of the elements $\mathrm{B},\ \mathrm{C},\ \mathrm{N}$ and F is

(A) $\mathrm{F} < \mathrm{N} > \mathrm{C} > \mathrm{B}$

(B) $\mathrm{C} > \mathrm{B} > \mathrm{N} > \mathrm{F}$

(C) $\mathrm{B} > \mathrm{C} > \mathrm{N} > \mathrm{F}$

(D) $\mathrm{F} > \mathrm{N} > \mathrm{C} > \mathrm{B}$

Q.7

If $1.96 \times 10^4\ \mathrm{kJ}$ energy is supplied to $1\ \mathrm{mole\ Li}$ it ionises to Li^{3+}. If IE_1 for Li is $520\ \mathrm{kJ\ mol^{-1}}$ and IE for H is $2.18 \times 10^{-18}\mathrm{Jatom^{-1}}$. The IE_2 for Li will be

(A) $7270\ \mathrm{kJ\ mol^{-1}}$

(B) $11.81 \times 10^3\ \mathrm{kJ\ mol^{-1}}$

(C) $6.54 \times 10^{-3}\ \mathrm{kJ\ mol^{-1}}$

(D) $3.32 \times 10^4\ \mathrm{kJ\ mol^{-1}}$

Q.8

Identify the incorrect statement(s) among the following :

(A) If spin quantum number curve having 3 possible values then 4th period of periodic table would have maximum of 27 elements.

(B) Platinum (Pt) is one of the typical transition metal.

@IITJEE_Advanced

(C) total number of f - electrons in Gd (at. No. 64) is 8

(D) atomic radius of Ga is nearly equal to that of Al.

Q.9

Select equations having endothermic step :

(A) $\mathrm{S}^{-}(\mathrm{g}) \longrightarrow \mathrm{S}^{2-}(\mathrm{g})$

(B) $\mathrm{Na}^{+}(\mathrm{g})+\mathrm{Cl}^{-}(\mathrm{g}) \longrightarrow \mathrm{NaCl}\,(\mathrm{s})$

(C) $\mathrm{N}(\mathrm{g}) \longrightarrow \mathrm{N}^{-}(\mathrm{g})$

(D) $\mathrm{Al}^{2+}(\mathrm{g}) \longrightarrow \mathrm{Al}^{3+}(\mathrm{g})$

Q.10

Linus Pauling wanted to determine the electron gain enthalpy (electron affinity) of an unknown element $\mathbf{X}$. He took one mole of atoms of $\mathbf{X}$ and found that on absorption of $410\ \mathrm{kJ}$ of energy half of $\mathbf{X}$-atoms transfer one electron to the other half. To convert all the resulting $\mathbf{X}^{-}$ions to $\mathbf{X}^{+}$ions, an additional energy of $735\ \mathrm{kJ}$ was required. What was the electron gain enthalpy of $\mathbf{X}$ as obtained by Linus Pauling?

(A) $-270\ \mathrm{kJ\ mol}^{-1}$

(B) $-325\ \mathrm{kJ\ mol}^{-1}$

(C) $-540\ \mathrm{kJ\ mol}^{-1}$

(D) $-650\ \mathrm{kJ\ mol}^{-1}$

Q.11

For the gaseous reaction $\mathrm{K}+\mathrm{F} \rightarrow \mathrm{K}^{+}+\mathrm{F}^{-}, \Delta \mathrm{H}$ was calculated to be $19 \mathrm{kcal}$ under conditions where the cations and anions were prevented by electrostatic separation from combining with each other. The ionization energy of K is $4.3 \mathrm{eV}$. What is the electron affinity of F ?

(A) $2.3 \mathrm{eV}$

(B) $3.5 \mathrm{eV}$

(C) $4.3 \mathrm{ev}$

(D) $5.4 \mathrm{ev}$

Q.12

Paragraph:
Nuclear charge actually experienced by an electron is termed as effective nuclear charge. The effective nuclear charge Z^{*} actually depends on type of shell and orbital in which electron is actually present. The relative extent to which the various orbitals penetrate the electron clouds of other orbitals is $\mathrm{s}>\mathrm{p}>\mathrm{d}>\mathrm{f}$ (for the same value of n)
The phenomenon in which penultimate shell electrons act as screen or shield in between nucleus and valence shell electrons and thereby reducing nuclear charge is known as shielding effect. The penultimate shell electrons repel the

@IITJEE_Advanced

valence shell electron to keep them loosely held with nucleus. It is thus evident that more is the shielding effect, lesser is the effective nuclear charge and lesser is the ionization energy.

Question:

Which of the following valence electron experience maximum effective nuclear charge ?

(A) $4\ s^1$

(B) $4p^1$

(C) $3\ d^1$

(D) $2p^3$

Q.13

Paragraph:

Nuclear charge actually experienced by an electron is termed as effective nuclear charge. The effective nuclear charge Z^* actually depends on type of shell and orbital in which electron is actually present. The relative extent to which the various orbitals penetrate the electron clouds of other orbitals is $s > p > d > f$ (for the same value of n)

The phenomenon in which penultimate shell electrons act as screen or shield in between nucleus and valence shell electrons and thereby reducing nuclear charge is known as shielding effect. The penultimate shell electrons repel the valence shell electron to keep them loosely held with nucleus. It is thus evident that more is the shielding effect, lesser is the effective nuclear charge and lesser is the ionization energy.

Question:

Which of the following is not concerned to effective nuclear charge ?

(A) Higher ionization potential of carbon than boron

(B) Higher ionization potential of magnesium than aluminium

(C) Higher values of successive ionization energy

(D) Higher electronegativity of higher oxidation state

Q.14

Paragraph:

Nuclear charge actually experienced by an electron is termed as effective nuclear charge. The effective nuclear charge Z^* actually depends on type of shell and orbital in which electron is actually present. The relative extent to which the various orbitals penetrate the electron clouds of other orbitals is $s > p > d > f$ (for the same value of n)

The phenomenon in which penultimate shell electrons act as screen or shield in between nucleus and valence shell electrons and thereby reducing nuclear charge is known as shielding effect. The penultimate shell electrons repel the valence shell electron to keep them loosely held with nucleus. It is thus evident that more is the shielding effect, lesser is the effective nuclear charge and lesser is the ionization energy.

Question:

Ionization energy is not influenced by -

(A) Size of atom

(B) Effective nuclear charge

@IITJEE_Advanced

(C) Electrons present in inner shell

(D) Change in entropy

Q.15

Paragraph:

Numerous forms of the periodic table have been revised from time to time. A modern version, which is most convenient and widely used is the long or extended form of periodic table. The Aufbau principle (electrons are filled in the progressive order of their increasing energy, i.e. by $n + \ell$ rule) and the electronic configuration of atom provide a theoretical foundation for the periodic classification. The horizontal rows are called periods. There are altogether seven periods. The first period consists of 2 elements. The subsequent periods consist of $8, 8, 18, 18$ and 32 elements respectively. The seventh period is incomplete and like the sixth period would have maximum of 32 elements.

Elements having similar outer electronic configurations in their atoms are grouped in vertical columns. These are referred to as groups or families. According to the recommendation of IUPAC, the groups are numbered 1 to 18 replacing the older notation of groups 0 , IA, IIA, ...VII A, VIII, IB , ... VII B.

Each successive period in the periodic table is associated with the filling up next higher principal energy level following aufbau sequence. The number of elements in each period is twice the number of atomic orbitals available in the energy level that is being filled. All the elements are classified, into four blocks, i.e., s-block, p-block, d-block and f-block depending on the type of atomic orbitals that are being filled with the last electron of the element.

Question:

The element with atomic number 56 is likely to have the same outer shell configuration as the element with, atomic number:

(A) 12

(B) 18

(C) 14

(D) None of these

Q.16

Paragraph:

Numerous forms of the periodic table have been revised from time to time. A modern version, which is most convenient and widely used is the long or extended form of periodic table. The Aufbau principle (electrons are filled in the progressive order of their increasing energy, i.e. by $n + \ell$ rule) and the electronic configuration of atom provide a theoretical foundation for the periodic classification. The horizontal rows are called periods. There are altogether seven periods. The first period consists of 2 elements. The subsequent periods consist of $8, 8, 18, 18$ and 32 elements respectively. The seventh period is incomplete and like the sixth period would have maximum of 32 elements.

Elements having similar outer electronic configurations in their atoms are grouped in vertical columns. These are referred to as groups or families. According to the recommendation of IUPAC, the groups are numbered 1 to 18 replacing the older notation of groups 0 , IA, IIA, ...VII A, VIII, IB , ... VII B.

Each successive period in the periodic table is associated with the filling up next higher principal energy level following aufbau sequence. The number of elements in each period is twice the number of atomic orbitals available in the energy level that is being filled. All the elements are classified, into four blocks, i.e., s-block, p-block, d-block

@IITJEE_Advanced

and f-block depending on the type of atomic orbitals that are being filled with the last electron of the element.

Question:

If aufbau rule is not followed, $\mathrm{Ca}-20$ will be placed in ..... block.

(A) s-

(B) p-

(C) d-

(D) f

Q.17

Paragraph:

Numerous forms of the periodic table have been revised from time to time. A modern version, which is most convenient and widely used is the long or extended form of periodic table. The Aufbau principle (electrons are filled in the progressive order of their increasing energy, i.e. by $\mathrm{n}+\ell$ rule) and the electronic configuration of atom provide a theoretical foundation for the periodic classification. The horizontal rows are called periods. There are altogether seven periods. The first period consists of 2 elements. The subsequent periods consist of $8, 8, 18, 18$ and 32 elements respectively. The seventh period is incomplete and like the sixth period would have maximum of 32 elements.

Elements having similar outer electronic configurations in their atoms are grouped in vertical columns. These are referred to as groups or families. According to the recommendation of IUPAC, the groups are numbered 1 to 18 replacing the older notation of groups 0 , IA, IIA, ...VII A, VIII, IB , ... VII B.

Each successive period in the periodic table is associated with the filling up next higher principal energy level following aufbau sequence. The number of elements in each period is twice the number of atomic orbitals available in the energy level that is being filled. All the elements are classified, into four blocks, i.e., s-block, p-block, d-block and f-block depending on the type of atomic orbitals that are being filled with the last electron of the element.

Question:

What is the position of the element in the periodic table satisfying the electronic configuration ($\mathrm{n}-1$) d^1 ns^2 for $\mathrm{n}=4$

(A) 3rd period and 3rd group

(B) 4th period and 4th group

(C) 3rd period and 2nd group

(D) 4th period and 3rd group

Q.18

Match the column -

Column I (Atomic number)	**Column II (Position in the periodic table)**
(A) 52	(p) s-block
(B) 56	(q) p-block
(C) 57	(r) d-block
(D) 60	(s) f-block

(A) A-q; B-p ; C- r ; D-s;

@IITJEE_Advanced

(B) A-p; B-r ; C- s ; D-s;

(C) A-s; B-r ; C-s ; D-s;

(D) A-s; B-p ; C- p ; D-s;

Q.19

The electronic configuration of an element is $1\ s^2, 2\ s^2 2p^6, 3\ s^2 3p^4$. The atomic number is of the element directly below it

Q.20

The energy needed to convert three moles of sodium atoms in the gaseous state to sodium ions is $14.85 \times 10^{x} J$. [Given : The ionization energy of sodium is $495\ kJ\ mol^{-1}$] What is the value of x

Answers & Solutions

Q.1 Answer:

All isoelectronic ions belong to same period of the periodic table.

Solution:

In the isoelectronic species, all isoelectronic anions belong to the same period and cations to the next period.

Q.2 Answer:

VXYW

Solution:

(i) Alkali metal carbonates do not decompose even at red hot $\rightarrow V$

(ii) Transition metal ions having unpaired d-electrons are coloured in aq. sol./compound $\rightarrow X$

(iii) In case of Kr **van der Waal's radius is considered, which is largest atomic radius** $\rightarrow Y$

(iv) Si forms only acidic oxide $SiO_2 \rightarrow W$

Q.3 Answer:

2.8

Solution:

I.E. $+$ **E.A.** $= 275 + 86 = 361 kcal mol^{-1} = 361 \times 4.184 = 1510.42\ kJ\ mol^{-1}$

$\therefore$ **Electronegativity** $= \frac{1510.42}{540} = 2.797 = 2.8$

Q.4 Answer:

after III B group and in the 6th period

Solution:

The actual position of Lanthanoids in the periodic table is at group number 3 and period number 6.

@IITJEE_Advanced

Q.5 Answer:

A decrease in the ionization potential

Solution:

Screening effect of inner electrons will lead to decrease in attraction of outer electrons from the nucleus. Hence there is decrese in ionization potential because it will be easier to remove outermost electron now.

Q.6 Answer:

$\mathrm{F} > \mathrm{N} > \mathrm{C} > \mathrm{B}$

Solution:

The correct order of non-metallic character of the elements is $\mathrm{F} > \mathrm{N} > \mathrm{C} > \mathrm{B}$. The given elements belongs to same period. On moving across a period atomic size decreases as electrons are being added to same shell, nuclear charge increases and the tendency to gain electrons increases. Thus, the non-metallic character also increases.

Q.7 Answer:

$7270\ \mathrm{kJ\ mol^{-1}}$

Solution:

$$\mathrm{Li_{(s)} \longrightarrow Li^{+3}_{(s)} + 3e^-} \quad \Delta \mathrm{H} = 1.96 \times 10^4\ \mathrm{kJ\ mol^{-1}}$$
$$\mathrm{Li_{(s)} \longrightarrow Li^{+}_{(s)} + e^-} \quad \mathrm{IE_1} = 520\ \mathrm{kJ\ mol^{-1}}$$
$$\mathrm{Li^{+}_{(s)} \longrightarrow Li^{+2}_{(s)} + e^-} \quad \mathrm{IE_2} = a\ \mathrm{kJ\ mol^{-1}}$$
$$\mathrm{Li^{+2}_{(s)} \longrightarrow Li^{+3}_{(s)} + e^-} \quad \mathrm{IE_3} = b\ \mathrm{kJ\ mol^{-1}}$$

also $b = \mathrm{E_1}$ for $Li^{+2} = E_1$ for $H \times Z^2$

(as Li^{2+} is a single electron species bohr's model is applicable

$$= 2.18 \times 10^{-18} \times 3^2 \times 6.023 \times 10^{23}\ \mathrm{J\ mol^{-1}}$$
$$= 11.81 \times 10^{46}\ \mathrm{J\ mol^{-1}} = 11.81 \times 10^3\ \mathrm{kJ\ mol^{-1}}$$

For the above eq

$1.96 \times 10^4 = 520 + a + 11.81 \times 10^3$

$\mathrm{a} = 7270\ \mathrm{kJ\ mol^{-1}}$

Q.8 Answer:

total number of f - electrons in Gd (at. No. 64) is 8

Solution:

(A) Three possible spin quantum number means an orbital can have at the most three electrons. So 4th period involves, $3\ \mathrm{s}$, $3\ \mathrm{d}$ and $4\mathrm{p}$ subshells.

So, maximum of electrons (and hence elements) $= 4s^3 3d^{15} 4p^9 = 3 + 15 + 9 = 27$

(B) All elements of VIII th group (i.e. $8, 9, 10$ groups) are called typical transition metals. Pt also belongs to 10th group. So it is a typical transition metal.

(C) E.C. of $\mathrm{Gd}(64) = [Xe]4\mathrm{f}^7 5d^1 6s^2$ so number of $\mathrm{f}-$ electrons $= 7$

(D) Due to ineffective shielding by $3\ \mathrm{d}$ - electrons in Ga, size of Ga is smaller (almost equal) to that of Al. Hence (C) is correct.

@IITJEE_Advanced

Q.9 Answer:

$S^{-}(g) \longrightarrow S^{2-}(g)$

Solution:

(A) $S^{-}(g) \longrightarrow S^{2-}(g); \quad \Delta H_{e.g.} = (+)$ **ve**

(B) $Na^{+}(g) + Cl^{-}(g) \longrightarrow NaCl(s); \quad \Delta H_{I.E.} = (-)ve$

$(C) N(g) \longrightarrow N^{-}(g); \quad \Delta H_{e.g.} = (+)ve$

(D) The reaction $Al^{2+}(g) \rightarrow Al^{3+}(g)$ **represents an endothermic process as energy is required to remove an electron from dipositve ion.**

It represents third ionization potential of Al.

Q.10 Answer:

-325 kJ mol^{-1}

Solution:

$\frac{1}{2}X \longrightarrow \frac{1}{2}X^{+} + e; \frac{1}{2}I.E \ldots (a)$

$e + \frac{1}{2}X \longrightarrow \frac{1}{2}X^{-}; \frac{1}{2}\Delta H_{eg} \ldots..(b)$

$X \longrightarrow \frac{1}{2}X^{+} + \frac{1}{2}X^{-}; \frac{1}{2}(I.E. + \Delta H_{eg}) = 410 \text{ kJ}$

by (a) + (b),

$I.E. + \Delta H_{eg} = 820 \text{ kJ} \quad \ldots (i)$

$\frac{1}{2}X^{-} \longrightarrow \frac{1}{2}X^{+} + 2e; \Delta H = 735 \text{ kJ} \ldots.. (c)$

(a)-(b) = (c)

$\frac{1}{2}I.E. - \frac{1}{2}\Delta H_{eg} = 735$

$\Rightarrow I.E. - \Delta H_{eg} = 1470 \quad \ldots (ii)$

solving (i) and (ii), we get,

$-2\Delta H_{eg} = 650 \Rightarrow \Delta H_{eg} = -325 \text{kJmol}^{-1}$

Q.11 Answer:

3.5eV

Solution:

$\Delta H = \left(\frac{19 \text{kcal}}{\text{mol}}\right)\left(\frac{4.18 \times 10^{3} \text{ J}}{\text{kcal}}\right)\left(\frac{1 \text{eV}}{1.60 \times 10^{-19} \text{ J}}\right)\left(\frac{1 \text{ mol}}{6.02 \times 10^{23} \text{atoms}}\right) = 0.82 \text{eV}$

$K(g) + F(g) \longrightarrow F^{-}(g) + K^{+}(g) \quad \Delta H = 19 \text{kcal} = 0.82 \text{eV}$

$K(g) \longrightarrow K^{+}(g) + e^{-} \; F(g) + e^{-} \longrightarrow F^{-}(g)$

$IE = 4.3 \text{eV}$

$EA = \Delta H - IE = 4.3 - 0.82 = 3.5 \text{eV}$

Q.12 Answer:

$2p^{3}$

@IITJEE_Advanced

Solution:

(i) Electrons closer to nucleus will experience higher effective nuclear change.
$2p^3$ is closer to nucleus than $4\ s^1$ as principal quantum number is concerned first.
(ii) Magnesium having higher ionization potential due to more stable electronic arrangement [Ne] $3\ s^2$ in comparison to aluminium [Ne] $3\ s^2 3p^1$.
(iii) Ionization energy is not affected by entropy.

Q.13 Answer:

Higher ionization potential of magnesium than aluminium

Solution:

(i) Electrons closer to nucleus will experience higher effective nuclear change.
$2p^3$ is closer to $4\ s^1$ as principal quantum number is concerned first.
(ii) Magnesium having higher ionization potential due to more stable electronic arrangement [Ne] $3\ s^2$ in comparison to aluminium [Ne] $3\ s^2 3p^1$.
(iii) Ionization energy is not affected by entropy.

Q.14 Answer:

Change in entropy

Solution:

(i) Electrons closer to nucleus will experience higher effective nuclear change.
$2p^3$ is closer to $4\ s^1$ as principal quantum number is concerned first.
(ii) Magnesium having higher ionization potential due to more stable electronic arrangement [Ne] $3\ s^2$ in comparison to aluminium [Ne] $3\ s^2 3p^1$.
(iii) Ionization energy is not affected by entropy.

Q.15 Answer:

12

Solution:

Electronic configuration of element with $Z = 56; [Xe]^{54} 6\ s^2$
Electronic configuration of element with

$$Z = 12; [Ne]^{10} 3\ s^2$$

As both have same valence shell electron configurations they belong to s-block and and same group i.e. 2nd, the alkaline earth metals.
If aufbau rule is not followed then $3\ d$ will be filled before $4\ s$. So last electron of $Ca - 20$ will go to $3\ d$-subshell. Hence the element will belong to d-block inplace of s-block.
When $n = 4$, the configuration will be $[Ar]^{18} 3\ d^1 4\ s^2$ and thus period is fourth and group number is third, $(1 + 2 = 3)$.

@IITJEE_Advanced

Q.16 Answer:

d-

Solution:

Electronic configuration of element with $Z = 56; [Xe]^{54} 6\ s^2$

Electronic configuration of element with

$$Z = 12; [Ne]^{10} 3\ s^2$$

As both have same valence shell electron configurations they belong to s-block and and same group i.e. 2nd, the alkaline earth metals.

If aufbau rule is not followed then $3\ d$ **will be filled before** $4\ s$**. So last electron of** $Ca - 20$ **will go to** $3\ d$**-subshell. Hence the element will belong to d-block inplace of s-block.**

When $n = 4$**, the configuration will be** $[Ar]^{18} 3\ d^1 4\ s^2$ **and thus period is fourth and group number is third,** $(1 + 2 = 3)$**.**

Q.17 Answer:

4th period and 3rd group

Solution:

Electronic configuration of element with $Z = 56; [Xe]^{54} 6\ s^2$

Electronic configuration of element with

$$Z = 12; [Ne]^{10} 3\ s^2$$

As both have same valence shell electron configurations they belong to s-block and and same group i.e. 2nd, the alkaline earth metals.

If aufbau rule is not followed then $3\ d$ **will be filled before** $4\ s$**. So last electron of** $Ca - 20$ **will go to** $3\ d$**-subshell. Hence the element will belong to d-block inplace of s-block.**

When $n = 4$**, the configuration will be** $[Ar]^{18} 3\ d^1 4\ s^2$ **and thus period is fourth and group number is third,** $(1 + 2 = 3)$**.**

Q.18 Answer:

A-q; B-p ; C- r ; D-s;

Solution:

(A) $-\mathbf{q}, (\mathbf{B}) - \mathbf{p}, (\mathbf{C}) - \mathbf{r}, (\mathbf{D}) - \mathbf{s}.$

$52[Kr]^{36} 4\ d^{10} 5\ s^2 5p^4 \Rightarrow$ **p-block**

$56[Xe]^{54} 6\ s^2 \quad \Rightarrow s - block$

$57[Xe]^{54} 5\ d^1 6\ s^2 \quad \Rightarrow d - block$

Q.19 Answer:

—

Solution:

@IITJEE_Advanced

At. No. $= 16(\mathrm{S})$

Next element below this element has atomic number $= 16 + 18 = 34$

Q.20 Answer:

—

Solution:

Energy needed to convert 1 mole of sodium (g) **to sodium ions** $= 495\ \mathrm{kJ}$

$\therefore$ **Energy needed to convert 3 mole of** $\mathrm{Na(g)}$ **to** $\mathrm{Na^+}$ **ions** $= 495 \times 3 = 1485\ \mathrm{kJ}$

So $x = 5$

@IITJEE_Advanced